

Asymptotic Matching [B0] Pg 348 Sec 7.4

Intro: will be used quite often in boundary layer theory and WKB theory (ch 9 and ch 10 respectively)

Principle of asymptotic matching:

① Identify regions: the interval is broken into 2 or more overlapping subintervals.

(different behavior of x or y coupled with ε is the reason to divide our interval into different regions)

② Apply perturbation theory on each region.

③ Match those solutions in the overlapping region.

Boundary Layer Theory [Bo] Pg 431 ch 9

Def. boundary layer: the narrow region where the solution of a differential equation changes rapidly.

Note: basically we consider perturbing parameter ε appeared as the coefficient of ODEs' highest order term.

property: as $\varepsilon \rightarrow 0$, the thickness of boundary layer should $\rightarrow 0$

Simplified situation: Now we only consider isolated (well-separated) boundary layer

Note: if that property doesn't hold, then that region is not typically boundary layer (doesn't vanish as $\varepsilon \rightarrow 0$)
those solved in [Bo] ch 10, WKB

inner region: boundary layer region.

outer region: region of slow variation of solution.

boundary layer theory: collection of perturbation methods solving differential equations whose solutions exhibit boundary layer structure.

roughly process:

check: $\varepsilon \rightarrow 0$ regular.

- ① Identify boundary layers: just use regular expansion, then check the solution may have crazy behavior near some points, which are the locations of boundary layers.
- ② for those regions far away from boundary layers are outer regions, just use regular expansion by perturbation theory.
- ③ for the regions near boundary layers are inner regions, these intervals are pretty small, we can take limit to simplify the problem, solve differential equation easily

④ there will be some constants generated during the process ② and ③, so we just using our boundary conditions, as well as asymptotic matching in the overlapping regions to determine them.

⑤ the same in the process ④, asymptotic matching can form $y_{\text{uniform}} = y_{\text{inner}} + y_{\text{outer}} - y_{\text{matching}}$